Bayesian SC vs Bayesian SDID

Geo-Lift Comparison on Simulated Panel

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June 18, 2026

1 What this project does

Inspired by a real-world migration in media-measurement platforms from Bayesian Synthetic Control (BSC) toward Bayesian Synthetic Difference-in-Differences (BSDID), this project fits both models on the same simulated panel with a known treatment effect and reports the difference. Results are shown for one representative simulated panel; a full Monte Carlo study would be needed to separate systematic bias from sampling noise.

2 Data

A 21-unit, 40-week panel. One treated unit, 20 controls. The treatment hits the treated unit at week 31 with a true per-week effect of $\tau = 25{,}000$ SEK. Each unit has its own baseline, a shared trend, annual seasonality, and Gaussian noise.

3 Model specification

3.1 Notation

Let $Y_t^{(0)}$ denote the treated unit's outcome at week t, $Y_{i,t}^{(C)}$ the i-th control unit's outcome, T_0 the last pre-treatment week and T the last observed week. Let $\omega \in \Delta^{N-1}$ (the N-simplex) be unit weights and $\lambda \in \Delta^{T_0-1}$ time weights.

3.2 Bayesian Synthetic Control (BSC)

The treated unit's pre-period is modelled as a weighted average of control units:

$$Y_t^{(0)} \sim \mathcal{N}\left(\sum_{i=1}^{N} \omega_i \, Y_{i,t}^{(C)},\ \sigma^2\right), \quad t = 1, \ldots, T_0$$

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}_{\text{emp}}^2)$$

where $\widehat{\sigma}_{\text{emp}} = \text{sd}(Y_{1:T_0}^{(0)})$ is the empirical pre-period standard deviation of the treated unit.

ATT (generated quantity): For each posterior draw s,

$$\text{counterfactual}_t^{(s)} = \sum_i \omega_i^{(s)} \, Y_{i,t}^{(C)}, \quad \text{ATT}_t^{(s)} = Y_t^{(0)} - \text{counterfactual}_t^{(s)}$$

for $t > T_0$. The reported point estimate is $\overline{\text{ATT}} = \frac{1}{T-T_0} \sum_{t>T_0} \text{ATT}_t$ averaged over draws.

Posterior:

$$p(\omega, \sigma \mid Y^{(0)}, Y^{(C)}) \ \propto\ \text{Dirichlet}(\omega \mid \mathbf{1}) \cdot \mathcal{N}^+(\sigma) \prod_{t=1}^{T_0} \mathcal{N}\left(Y_t^{(0)} \;\middle|\; \sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right)$$

3.3 Bayesian Synthetic Difference-in-Differences (BSDID)

Same ω-weighted control layer as BSC, plus a second simplex λ of pre-period time weights. The ATT subtracts the λ-weighted pre-period gap:

$$\mathrm{ATT}_t = \left(Y_t^{(0)} - \sum_i \omega_i Y_{i,t}^{(C)}\right) - \sum_{s=1}^{T_0} \lambda_s \left(Y_s^{(0)} - \sum_i \omega_i Y_{i,s}^{(C)}\right)$$

Two likelihoods identify ω and λ jointly:

$$\text{Likelihood 1 (treated pre):} \quad Y_t^{(0)} \sim \mathcal{N}\left(\sum_i \omega_i Y_{i,t}^{(C)}, \sigma^2\right), \quad t = 1, \dots, T_0$$

$$\text{Likelihood 2 (controls' pre-vs-post):} \quad \overline{Y_{i,\text{post}}^{(C)}} \sim \mathcal{N}\left(\sum_t \lambda_t \, Y_{i,t}^{(C)},\ \sigma_\lambda^2\right), \quad i = 1, \dots, N$$

where $\overline{Y_{i,\text{post}}^{(C)}} = \frac{1}{T-T_0} \sum_{t>T_0} Y_{i,t}^{(C)}$.

Priors:

$$\omega \sim \text{Dirichlet}(\mathbf{1}_N), \quad \lambda \sim \text{Dirichlet}(\mathbf{1}_{T_0})$$
$$\sigma \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\text{emp}}), \quad \sigma_\lambda \sim \mathcal{N}^+(0,\, \widehat{\sigma}^2_{\lambda,\text{emp}})$$

where $\widehat{\sigma}_{\lambda,\text{emp}}$ is the empirical standard deviation of $\overline{Y_{i,\text{post}}^{(C)}}$ across controls.

Posterior:

$$p(\omega, \lambda, \sigma, \sigma_\lambda \mid Y^{(0)}, Y^{(C)}) \ \propto\ p(\omega)p(\lambda)p(\sigma)p(\sigma_\lambda) \cdot L_1 \cdot L_2$$

with L_1 and L_2 the two likelihoods above.

4 Inference

Both models fitted with Stan via cmdstanpy using HMC-NUTS. Diagnostics below.

Metric	**Value**
Chains	4
Warmup iterations	1 500 per chain
Sampling iterations	1 500 per chain
Post-warmup draws	6 000
max $\widehat{R}$	≈ 1.00
Divergences	0

Table 1: Convergence diagnostics for both BSC and BSDID fits.

5 Results

	Truth	**BSC**	**BSDID**
Mean ATT	25 000	19 550	29 333
90% interval	–	[14 386, 23 889]	[26 154, 32 301]
Interval width	–	9 503	**6 147**
|Error|	–	5 450	**4 333**

Table 2: ATT posterior summary on one simulated panel. BSDID has both a smaller absolute error and a 35% tighter interval than BSC. A Monte Carlo study over many panels would be required to separate systematic bias from sampling noise.

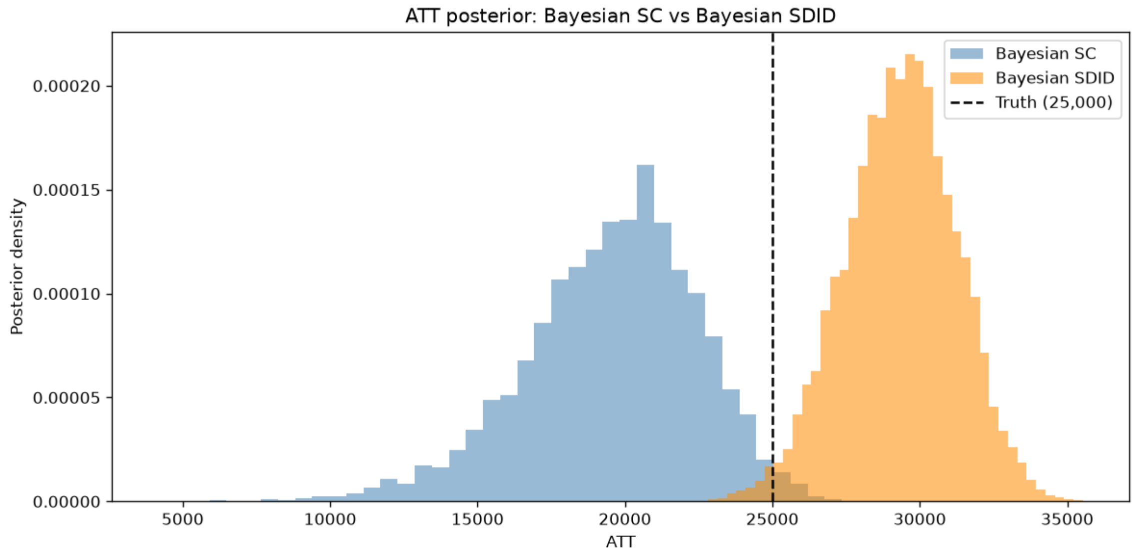

Figure 1: ATT posteriors. Dashed line is truth.

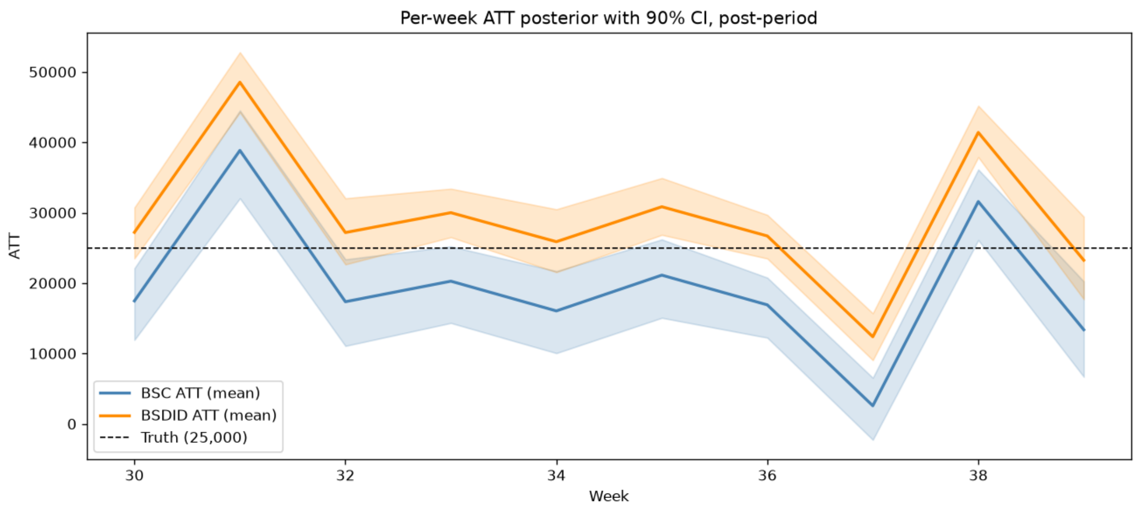

Figure 2: Per-week ATT, posterior mean and 90 % CI.

6 Why BSDID can outperform BSC

The treated unit sits slightly below its synthetic control in the pre-period. BSC reads that offset as part of the post-period treatment effect and underestimates τ by 22 %. BSDID subtracts the offset out of the ATT formula. On this panel it overshoots by 17 %, but with a tighter posterior and a smaller absolute error.

This is consistent with the motivation behind Arkhangelsky et al.'s SDID framework. It matters whenever the treated region happens to be a little above or below the convex hull of its controls in the pre-period, which is the common case on real geo data.

7 Limitations

The panel is synthetic and cleaner than reality. The simulator produces observations from a fixed data-generating process with additive Gaussian noise, a stable trend, and clean annual seasonality. Real geo data has non-stationarity, structural breaks, inter-region spillovers, correlated shocks across regions during macro events, and heteroskedastic noise. Signal-to-noise ratio here is unrealistically high, so absolute numbers (35 % tighter interval, 20 % lower error) reflect the synthetic setup and would move on real data.

Single seed. All results come from one random draw of the data-generating process. A full Monte Carlo study over many seeds would be needed to separate systematic bias from sampling noise.

Uniform Dirichlet priors. No smoothness regularization on λ beyond the Dirichlet, and no informative priors on ω. Production models typically use priors tied to geographic similarity, market size, or category benchmarks.

Pure on/off treatment. Only binary treatment is modelled. Real geo-lift experiments have varying spend intensities and treatment timing across regions.

No hierarchical pooling. A single experiment fitted in isolation. A mature system pools information across many past lift tests.